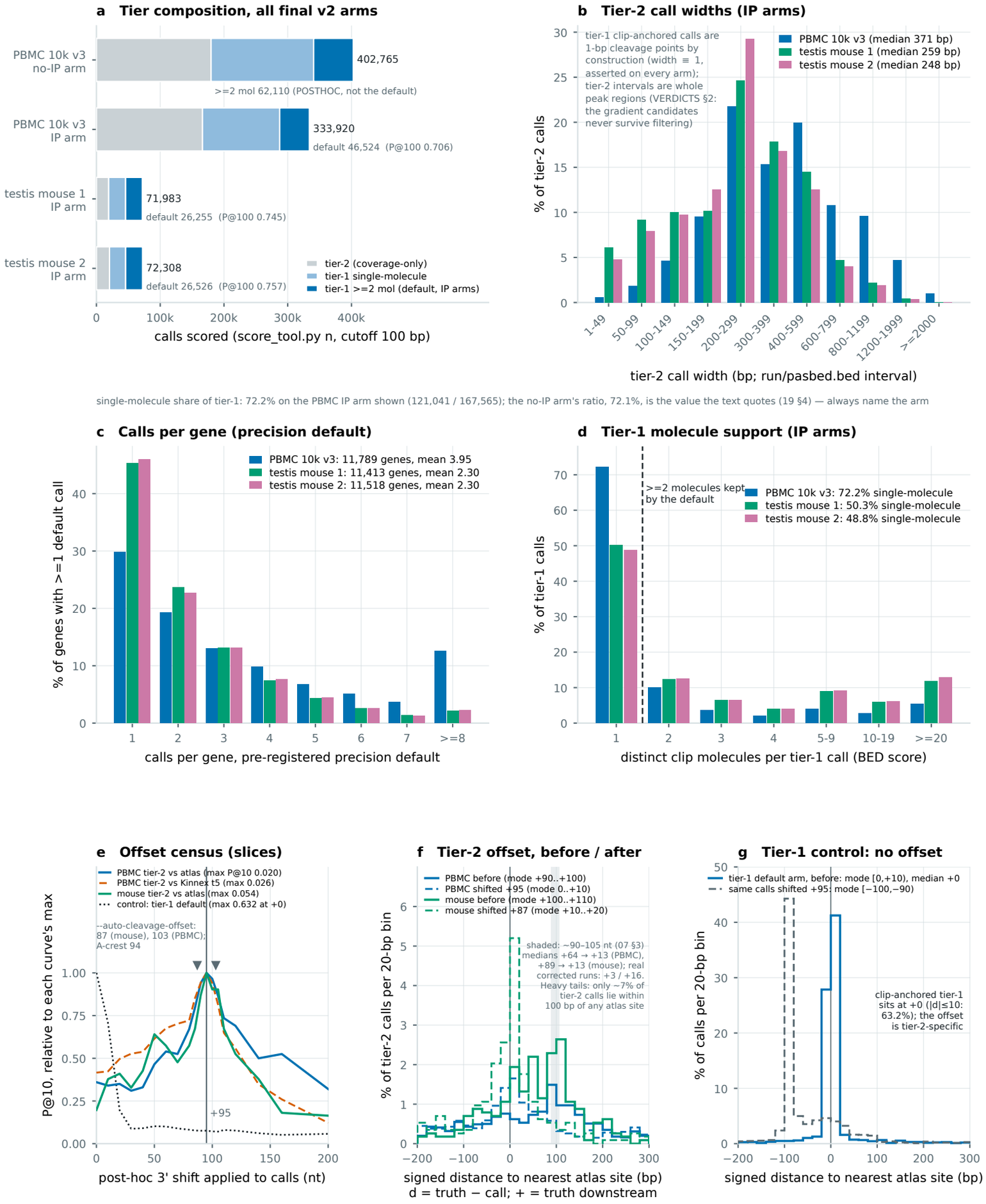



Offset correction does NOT rescue tier-2 (VERDICTS §3, FAIL on 28 §2): after --cleavage-offset 95, tier-2 atlas-agreement P@100 0.085 vs genic-shuffle null 0.028 (3.0×; was 2.2×); internal-priming decoy rate 0.020 → 0.065 (×3.2); R_det unchanged (0.0365 → 0.0369); mouse 0.179 vs null 0.016. The default arm is byte-identical across every offset arm (tier-1 exempt by design). Disposition: report inferred cleavage, do not move calls.

Fig S2 | v2 peak-call QC and the tier-2 cleavage-offset census. (a) Tier composition of every final v2 arm (code 9dfdefb), scored n from the single scoring path (score_tool.py, cutoff 100 bp): the pre-registered precision default (tier-1, IP filter, >=2 distinct clip molecules) keeps 46,524 of 167,565 tier-1 PBMC calls (P@100 0.7062); the no-IP >=2-molecule file is POSTHOC and is not the default. (b) Tier-2 calls are wide intervals (whole peak regions); tier-1 calls are 1-bp clip-anchored cleavage points. (c) Default calls per gene. (d) Tier-1 distinct-clip-molecule distributions; the >=2 threshold is the pre-registered default cut (12 CORRECTION). (e-g) The ~90-105 nt tier-2 coverage offset (predicted in 07 §3: 10x R2 coverage runs out before the poly(A) junction; measured in the Lead-B census on the development slices, PBMC chr19+21 / mouse1 chr18+19). Four independent estimates agree: tool --auto-cleavage-offset 103 (PBMC) / 87 (mouse), and post-hoc argmax +95 on atlas P@10 (both species) and on Kinnex t5 P@10 (PBMC; t5 = >=5 long-read records at the terminus, not de-duplicated UMIs). The signed-distance mode moves to ~0 when corrected (f) and the clip-anchored tier-1 control sits at +0 (g) -- the offset is real and tier-2-specific. It is also a negative result: correction fails every pre-registered acceptance criterion of 28 §2 (boxed verdict). Sources: 19 + final_v2_verify (FIXED) score TSVs; v2 run outputs (read-only); results/paramsweep/ arms_all.tsv, leadB/, VERDICTS.md §2-§3 (slice sweep executing 28 §3 step 1; provisional pending verifier). Slice caveat: the PBMC slice is clip-richer than the genome (paramsweep README §1); whether the offset holds for all 166,355 full-BAM tier-2 sites is an inference from the slices, not a measurement. Every plotted value: results/figures/manuscript/figS2_peakqc_*.tsv.